Curriculum Vitae

Masoud Goharimanesh, Ph.D.

Associate Professor of Mechanical Engineering

University of Torbat Heydarieh (UTH), Iran



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PERSONAL INFORMATION

Full Name:	Masoud Goharimanesh	Phone:	(+98) 915 358 6831
Marital Status	Married	Email:	m.goharimanesh@gmail.com / m.goharimanesh@torbath.ac.ir
Nationality:	Iranian		

+40 Journal Papers | **+30** Conference Proceedings | H-index: **19**

EDUCATION

2011–2016 | Ph.D.
Mechanical Engineering – System Dynamics and Control
Ferdowsi University of Mashhad, Iran

Thesis Title: "Backing Up Enhancement of Multi-Articulated Vehicles using Fuzzy Reinforcement Learning"
Supervisor: Prof. Ali Akbar Akbari
Advisor: Prof. Mohammad Bagher Naghibi Sistani

2008–2011 | M.Sc.
Automotive Engineering – Vehicle Dynamics
Iran University of Science and Technology, Tehran

Thesis Title: "Enhancement of Lateral Vehicle Dynamics Using Tire Inflation Pressure"
Supervisor: Prof. Behrooz Mashadi

2002–2006 | B.Sc.
Mechanical Engineering – Solid Mechanics
Islamic Azad University, Mashhad Branch

CAREER TIMELINE

2016 | Ph.D. Completion
Ferdowsi University of Mashhad, Iran
Mechanical Engineering – System Dynamics and Control

2018 | Assistant Professor
University of Torbat Heydarieh (UTH), Iran
Mechanical Engineering Department

2019 | Erasmus+ Scholarship
Vilnius Gediminas Technical University (VGTU), Lithuania
Invited International Lecturer

2022 | Research Grant ★
Toronto Metropolitan University (TMU), Canada
Research Collaboration & Grant

2023 | Associate Professor
University of Torbat Heydarieh (UTH), Iran
Mechanical Engineering Department

RESEARCH INTERESTS



AWARDS & HONORS



Complete List of Journal Publications (with Hyperlinks)

2025–2026

1. **M. Goharimanesh**, S. Garg, S. Sajjadi, F. Janabi-Sharifi. "Autonomous control of soft robots using safe reinforcement learning and covariance matrix adaptation." *Engineering Applications of Artificial Intelligence*, 153, 110791, 2026.
2. S.M. Javadpour, M. Dehghani, M.M. Naserian, **M. Goharimanesh**. "Multiobjective Optimization of Photovoltaic–Thermal Collectors Using Nanofluids: Integrating Taguchi Method and Genetic Algorithm." *International Journal of Energy Research*, 2025 (1), 9967015, 2025. DOI:10.1155/er/9967015
3. E.A. Jannatabadi, **M. Goharimanesh**. "Scientific Fuzzy Machine Learning."

2024

4. M.R. Gharib, A. Koochi, M.R.S. Kolahi, **M. Goharimanesh**. "Dynamic modeling of three-finger micro-tweezers for grasping microparticles." *Computers and Electrical Engineering*, 118, 109453, 2024.
5. S.E. Razavi, H. Khodadadi, **M. Goharimanesh**. "Adjusting laser power to control the heat generated by nanoparticles at the site of a patient's cells." *IET Systems Biology*, 18(4), 119-128, 2024. DOI:10.1049/syb2.12093
6. A. Koochi, **M. Goharimanesh**, M.R. Gharib. "Investigation of surface roughness effects on the dynamic performance of electromagnetic nano-resonators." *Physica Scripta*, 99(6), 065511, 2024.
7. H. Namadchian, **M. Goharimanesh**, I. Zahmatkesh. "Multi-objective hydrothermal optimization of nanoparticles shape in nanofluid flow through an annular pipe having metal foam fins." *Numerical Heat Transfer, Part A: Applications*, 85(1), 21-41, 2024. DOI:10.1080/10407782.2023.2178986

2023

8. M.R. Gharib, A. Koochi, A. Heydari, **M. Goharimanesh**. "Robust qft control design for a multivariable torsion/bending of rotational nanoscanner." *ISA Transactions*, 142, 164-176, 2023.
9. A. Heydari, **M. Goharimanesh**, M.R. Gharib. "Dynamic viscosity analysis of hybrid nanofluid MWCNT-Al_2O_3/engine oil using statistical models with evaluating its performance in a double tube heat exchanger." *Journal of Thermal Analysis and Calorimetry*, 148(16), 8025-8039, 2023. DOI:10.1007/s10973-022-11608-w
10. A. Koochi, **M. Goharimanesh**, M.R. Gharib. "An improved fuzzy controller on electromechanical nano-tweezers." *Journal of Vibration and Control*, 29(11-12), 2658-2670, 2023. DOI:10.1177/10775463221083803
11. A. Mehrkish, F. Janabi-Sharifi, **M. Goharimanesh**, S. Norouzi-Ghazbi. "Multiple aspects grasp quality evaluation in underactuated grasp of tendon-driven continuum robots." *Intelligent Service Robotics*, 16(1), 33, 2023. DOI:10.1007/s11370-022-00449-z

2022

12. **M. Goharimanesh**, E.A. Jannatabadi, M. Dehghani, S.M. Javadpour. "Geometric and thermo hydrodynamic investigation of a 3D converging-diverging channel by Taguchi and ANFIS methods." *International Communications in Heat and Mass Transfer*, 138, 106285, 2022.
13. M. Golmohammadi, **M. Goharimanesh**, A. Afsharfard, M.R. Gharib. "Suppressing vibration of structures using nonlinear single unit impact dampers via Taguchi method." *Noise & Vibration Worldwide*, 53(9-10), 454-463, 2022. DOI:10.1177/09574565221128065
14. E.A.J. Abadi, H. Sahu, S.M. Javadpour, **M. Goharimanesh**. "Interpretable machine learning for developing high-performance organic solar cells." *Materials Today Energy*, 25, 100969, 2022.
15. A. Gharib, **M. Goharimanesh**, A. Koochi, M.R. Gharib. "Designing an evolutionary optimal washout filter based on genetic algorithm." *Aviation*, 26(1), 54, 2022. DOI:10.3846/aviation.2022.16570

2021

16. H.A. Zar, **M. Goharimanesh**, F. Janabi-Sharifi. "Mathematical Modeling and Machine Learning for Force Estimation on a Planar Catheter."
17. M. Lagzian, S.E. Razavi, **M. Goharimanesh**. "Investigation on tumor cells growth by Taguchi method." *Biomedical Signal Processing and Control*, 77, 103734, 2022.

18. M.R. Gharib, N. Jamali, S.N. Chamanabad, **M. Goharimanesh**. "Examining the role of empowerment criteria on employee performance: a quantitative analysis in the oil industry." *Journal of Engineering Management and Systems Engineering*, 2(2), 96-107, 2022.
19. A. Koochi, **M. Goharimanesh**, M.R. Gharib. "Nonlocal electromagnetic instability of carbon nanotube-based nano-sensor." *Mathematical Methods in the Applied Sciences*, 2021. DOI:10.1002/mma.7216
20. A. Koochi, **M. Goharimanesh**. "Nonlinear Oscillations of CNT Nano-resonator Based on Nonlocal Elasticity: The Energy Balance Method." *Reports in Mechanical Engineering*, 2(1), 41-50, 2021.
21. A. Marvi-Mashhadi, M.R. Sharifmoghadam, **M. Goharimanesh**, et al. "Methyl red biodegradation based on Taguchi method by two novel bacteria." *International Journal of Environmental Science and Technology*, 19, 1-12, 2021. DOI:10.1007/s13762-021-03264-8
22. E. Bahrami Jovein, A. Ghayouri Motlagh, **M. Goharimanesh**. "Estimation of Groundwater Flow Balance Components and Its Error Improvement Using Taguchi Method (Case Study: Mahvelat Aquifer)." *Iran-Water Resources Research*, 17(1), 195-211, 2021.

2020

23. **M. Goharimanesh**, A. Mehrkish, F. Janabi-Sharifi. "A Fuzzy Reinforcement Learning Approach for Continuum Robot Control." *Journal of Intelligent & Robotic Systems*, 100(3), 809-826, 2020. DOI:10.1007/s10846-020-01237-6 **Citations: 53**
24. S.M. Javadpour, E.A.J. Abadi, O.A. Akbari, **M. Goharimanesh**. "Optimization of geometry and nano-fluid properties on microchannel performance using Taguchi method and genetic algorithm." *International Communications in Heat and Mass Transfer*, 119, 104952, 2020.
25. R. Hekmat, **M. Goharimanesh**, B. Dadpour. "SAT-013 COMPARISON OF ARTIFICIAL NEURAL NETWORK MODEL AND ADAPTIVE NEURO-FUZZY INFERENCE SYSTEM (ANFIS) MODEL FOR PREDICTING HEMODIALYSIS INITIATION IN METHANOL POISONED PATIENTS." *Kidney International Reports*, 5(3), S6-S7, 2020.

2018–2019

26. A. Lashkaripour, **M. Goharimanesh**, A. Abouei-Mehrizi, D. Densmore. "An Adaptive Neural-Fuzzy Approach for Microfluidic Droplet Size Prediction." *Microelectronics Journal*, 78, 73-80, 2018. DOI:10.1016/j.mejo.2018.05.018
27. M. Rasouli, A.A. Mehrizi, **M. Goharimanesh**, A. Lashkaripour, S.R. Bazaz. "Multi-criteria optimization of curved and baffle-embedded micromixers for bio-applications." *Chemical Engineering and Processing-Process Intensification*, 132, 175-186, 2018.
28. A. Lashkaripour, A. Abouei Mehrizi, **M. Goharimanesh**, M.R. Rasouli, S.R. Bazaz. "Size-Controlled Droplet Generation in a Microfluidic Device for Rare DNA Amplification by Optimizing its Effective Parameters." *Journal of Mechanics in Medicine and Biology*, 18(1), 1850002, 2018. DOI:10.1142/S0219519418500021
29. J. Samavi, **M. Goharimanesh**, A. Akbari, E. Dezyani. "Optimisation of drilling parameters on St37 based on Taguchi method." *Journal of the Brazilian Society of Mechanical Sciences and Engineering*, 40(8), 1-9, 2018. DOI:10.1007/s40430-018-1245-y
30. D. Yousefi Kebria, M. Ghavami, S. Javadi, **M. Goharimanesh**. "Combining Experimental Study and ANFIS Modeling to Predict the Landfill Leachate Transport in Underlying Soil." *Environmental Monitoring and Assessment*, 190(1), 2018.
31. M. Zarif, A. Morvari, **M. Goharimanesh**. "Improvement Direct Power Control on PWM using Modified Deadbeat Controller." *International Journal of Control and Automation*, 11(9), 77-86, 2018.
32. **M. Goharimanesh**, E.A. Jannatabadi, A. Jahan, A. Akbari. "A decision-making methodology for material selection using genetic algorithm." *International Journal of Information and Decision Sciences*, 11(4), 269-299, 2019.

2017

33. **M. Goharimanesh**, A.A. Akbari. "Improving Lateral Dynamics of Vehicle Using Direct Yaw Moment Controller Based on Quantitative Feedback Theory." *Scientia Iranica*, 24(2), 662-672, 2017.
34. J.A. Esfahani, M.R. Safaei, **M. Goharimanesh**, L.R. De Oliveira, M. Goodarzi, et al. "Comparison of Experimental Data, Modelling and non-linear Regression on Transport Properties of Mineral Oil Based Nanofluids." *Powder Technology*, 317, 458-470, 2017. **Citations: 98**
35. **M. Goharimanesh**, E.A. Jannatabadi, H. Moeinkhah, M.B. Naghibi-Sistani, A.A. Akbari. "An intelligent controller for ionic polymer metal composites using optimized fuzzy reinforcement learning." *Journal of Intelligent & Fuzzy Systems*, 33(1), 125-136, 2017. DOI:10.3233/JIFS-161211

36. S.M. Javid, M. Passandideh-Fard, A. Faezian, **M. Goharimanesh**. "Slug and bubble flows in a flat sheet ultrafiltration module: experiments and numerical simulation." *International Journal of Multiphase Flow*, 91, 39-50, 2017.
37. A. Zolfaghari, **M. Goharimanesh**, A.A. Akbari. "Optimum design of straight bevel gears pair using evolutionary algorithms." *Journal of the Brazilian Society of Mechanical Sciences and Engineering*, 39(6), 2121, 2017. DOI:10.1007/s40430-017-0733-9
38. **M. Goharimanesh**, A.A. Akbari. "Vehicle dynamics improvement using developed robust controller." *Journal of Structures and Fluids*, 6(4), 2017. DOI:10.22044/JSFM.2017.920
39. **M. Goharimanesh**, S.M. Javid, S. Razavi-Bazaz, H. Rostami. "Reducing ice accretion using design of experiments based on Taguchi method." *Journal of Applied Science and Engineering*, 20(2), 165-172, 2017.

2015–2016

40. **M. Goharimanesh**, A. Lashkaripour, A.A. Akbari. "Optimization of Bio-Diesel Production using Multi-Objective Genetic Algorithms." *Journal of Applied Science and Engineering*, 19(2), 117-124, 2016.
41. **M. Goharimanesh**, A.A. Akbari, M.B. Naghibi Sistani. "Combining the principles of fuzzy logic and reinforcement learning for control of dynamic systems." *Journal of Applied and Computational Sciences in Mechanics*, 27(1), 103-116, 2016.
42. **M. Goharimanesh**, A.A. Akbari. "Optimum Parameters of Nonlinear Integrator using Design of Experiments Based on Taguchi Method." *Journal of Computational Applied Mechanics*, 46(2), 233-241, 2015.
43. **M. Goharimanesh**, A. Lashkaripour, A.A. Akbari. "A Comparison of Fuzzy Type 1 and 2 in Diabetics Control Based on Augmented Minimal Model." *Journal of World's Electrical Engineering and Technology*, 4(2), 70-75, 2015.
44. **M. Goharimanesh**, A. Lashkaripour, A. Abouei Mehrizi. "Fractional Order PID Controller for Diabetes Patients." *Journal of Computational Applied Mechanics*, 46(1), 69-76, 2015.
45. Mo. Goharimanesh, F. Ghasemzadeh, **M. Goharimanesh**. "Influential Feeding Dynamics of Sharks and Mathematical Formulas to Estimate the Power of Biting by Morphological Data." *Journal of Progress in Biological Sciences*, 5(2), 195-205, 2015.
46. **M. Goharimanesh**, A.A. Akbari. "On Using Fuzzy Reinforcement Learning to Control the Cancer Cells." *Journal of Cellular Immunotherapy*, 1(1), 34-35, 2015.
47. E. Abbasi Jannatabad, S. Akhlaghi Amiri, **M. Goharimanesh**, A.A. Akbari. "Vehicle Model Recognition Using Image Processing and Wavelet Analysis." *International Journal On Smart Sensing and Intelligent Systems*, 8(4), 2212-2230, 2015.

2014

48. **M. Goharimanesh**, A. Lashkaripour, S. Shariatnia, A.A. Akbari. "Diabetic Control using Genetic Fuzzy-PI Controller." *International Journal of Fuzzy Systems*, 16(2), 133-139, 2014. **Citations: 36**
49. **M. Goharimanesh**, A.A. Akbari, A.R. Akbarzadeh-T. "More Efficiency in Fuel Consumption Using Gearbox Optimization Based on Taguchi Method." *Journal of Industrial Engineering International*, 10(2), 1-8, 2014. DOI:10.1007/s40092-014-0061-y **Citations: 36**

Selected Conference Proceedings

Year	Conference	Title	Type
2020	ISME20, Tehran	Multi-Objective Optimization of a Three Degree Manipulator Controller	Oral
2015	IEEE AIM	Investigation of Changing of Center of Gravity in Quadrotor Maneuvers	Oral
2014	AVEC, Tokyo, Japan	Yaw Moment Control Using Fuzzy Reinforcement Learning	Oral
2010	ASME ESDA, Istanbul	Quantitative Feedback Theory Controller Design for Vehicle Stability	Oral

Journal Reviewer Experience

- International Journal of Fuzzy Systems
- International Journal of System Sciences
- Journal of Applied Mechanics
- Journal of Structures and Fluids

Technical Skills

PYTHON
MATLAB
ADAMS/CAR
CARSIM / TRUCKSIM
ANSYS / ABAQUS
COMSOL
SOLIDWORKS

Software Toolboxes & Packages Created

Toolbox	Description	Year
GOHARIZER	Software for digitizing pictures and graphs	2012
CAR20	Vehicle dynamics toolbox in MATLAB	2013
QUADSIM	MATLAB toolbox for quadrotor simulation	2014

Teaching Experience:

- Advanced Robotics
- Advanced Engineering Mathematics
- Mechatronics 1,2
- Machinery Dynamics
- Measurement and Control Systems
- Research Methods and Design of Experiments